Name : Sameer Gupta Roll No : 252010036

Execute and analyze Linux commands

1.ifconfig

```
mca@mca-OptiPlex-3050-AIO:~$ ifconfig
enp1s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.18.36.12  netmask 255.255.255.0  broadcast 172.18.36.255
        inet6 fe80::da9e:f3ff:fe7c:7ea3  prefixlen 64  scopeid 0x20<link>
        ether d8:9e:f3:7c:7e:a3  txqueuelen 1000  (Ethernet)
        RX packets 2878903  bytes 4296785176 (4.2 GB)
        RX errors 0  dropped 3004  overruns 0  frame 0
        TX packets 487182  bytes 43338817 (43.3 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 2906  bytes 305191 (305.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2906  bytes 305191 (305.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

virbr0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.122.1  netmask 255.255.255.0  broadcast 192.168.122.255
        ether 52:54:00:4b:33:f5  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 31 overruns 0  carrier 0  collisions 0

wlp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.0.205  netmask 255.255.255.0  broadcast 192.168.0.255
        inet6 fe80::f27e:de60:3608:a54a  prefixlen 64  scopeid 0x20<link>
        ether 0c:54:15:1d:79:4b  txqueuelen 1000  (Ethernet)
        RX packets 7750  bytes 1218473 (1.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
```

2 Ping

```
mca@mca-OptiPlex-3050-AIO:~$ ping google.com
PING google.com (142.251.43.14) 56(84) bytes of data.
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=1 ttl=116 time=18.0 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=2 ttl=116 time=6.68 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=3 ttl=116 time=8.23 ms
64 bytes from tsa03s08-in-f14.1e100.net (142.251.43.14): icmp_seq=4 ttl=116 time=12.1 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 6.678/11.247/18.018/4.372 ms
mca@mca-OptiPlex-3050-AIO:~$
```

3 . traceroute

```
mca@mca-OptiPlex-3050-AIO:~$ traceroute google.com
traceroute to google.com (142.251.43.14), 30 hops max, 60 byte packets
 1  _gateway (172.18.36.1)  1.500 ms  1.430 ms  1.384 ms
 2  172.18.35.2 (172.18.35.2)  0.672 ms  0.640 ms  0.600 ms
 3  undefined.hostname.localhost (103.47.105.129)  0.563 ms  0.526 ms  0.485 ms
 4  10.50.130.41 (10.50.130.41)  0.792 ms  0.756 ms  0.717 ms
 5  10.50.156.29 (10.50.156.29)  2.035 ms  1.957 ms  1.905 ms
 6  * * *
 7  173.194.123.250 (173.194.123.250)  1.600 ms  1.550 ms  1.494 ms
 8  192.178.84.175 (192.178.84.175)  2.743 ms  4.219 ms  3.736 ms
 9  142.251.77.101 (142.251.77.101)  2.488 ms 142.251.77.99 (142.251.77.99)  2.301 ms 142.251.77.101 (142.251.77.101)  2.757 ms
10  tsa03s08-in-f14.1e100.net (142.251.43.14)  2.652 ms  2.230 ms  2.181 ms
mca@mca-OptiPlex-3050-AIO:~$
```

4.nsLookup

```
mca@mca-OptiPlex-3050-AIO:~$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.43.14
Name:   google.com
Address: 2404:6800:4009:805::200e

mca@mca-OptiPlex-3050-AIO:~$
```

5.netstat

```
mca@mca-OptiPlex-3050-AIO:~$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 192.168.122.1:53        0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:33060         0.0.0.0:*               LISTEN      -
tcp6       0      0 ::1:631                 :::*                    LISTEN      -
tcp6       0      0 :::22                   :::*                    LISTEN      -
udp        0      0 192.168.122.1:53        0.0.0.0:*                           -
udp        0      0 127.0.0.54:53           0.0.0.0:*                           -
udp        0      0 127.0.0.53:53           0.0.0.0:*                           -
udp        0      0 0.0.0.0:67              0.0.0.0:*                           -
udp        0      0 224.0.0.251:5353        0.0.0.0:*                           2583/chrome
udp        0      0 224.0.0.251:5353        0.0.0.0:*                           2583/chrome
udp        0      0 0.0.0.0:5353            0.0.0.0:*                           -
udp        0      0 0.0.0.0:43465           0.0.0.0:*                           -
udp6       0      0 :::5353                 :::*                                -
udp6       0      0 :::55445                :::*                                -
mca@mca-OptiPlex-3050-AIO:~$
```

file permission

```
mca@mca-OptiPlex-3050-AIO:~$ ls -l file1.txt
-rw-rw-r-- 1 mca mca 0 Sep 20  2025 file1.txt
mca@mca-OptiPlex-3050-AIO:~$
```

user group admin

```
mca@mca-OptiPlex-3050-AIO:~$ whoami
mca
mca@mca-OptiPlex-3050-AIO:~$ groups
```

```
mca@mca-OptiPlex-3050-AIO:~$ groups
mca adm cdrom sudo dip plugdev users lpadmin
```

```
mca@mca-OptiPlex-3050-AIO:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nolog
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/no
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
tss:x:105:105:TPM software stack,,,:/var/lib/tpm:/bin/false
```

```
mca@mca-OptiPlex-3050-AIO:~$ cat /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,mca
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:
man:x:12:
```

admin privilages

```
mca@mca-OptiPlex-3050-AIO:~$ sudo -l
Matching Defaults entries for mca on mca-OptiPlex-3050-AIO:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty

User mca may run the following commands on mca-OptiPlex-3050-AIO:
    (ALL : ALL) ALL
```